Self-Evolving Clinical Deep Review: Eliminating Recurring Methodological False Alarms without Compromising Objective Error Recall via Comment-Driven Dual-Track Memory

Heurion Research Team
Correspondence: `research@heurion.ai`

November 2026

Abstract

Automated peer review and scientific verification systems face a fundamental precision–recall dilemma in clinical medicine. On one hand, single-pass large language model (LLM) reviewers and text-only multi-agent pipelines frequently miss objective numerical contradictions between manuscript prose and multi-column tables, inverted or out-of-bounds 95% confidence intervals, and intra-table row/cell arithmetic errors. Conversely, deterministic structural auditors alone struggle to capture open-ended semantic spin bias ($p \geq 0.05$ framed as clinical efficacy), while hybrid AST+LLM pipelines trigger recurring false alarms on legitimate study-design denominator transitions—such as intention-to-treat (ITT) randomized sample sizes versus per-protocol completers in CONSORT trials, pre- versus post-propensity-matching cohorts in STROBE observational studies, or dual parametric/non-parametric reporting across skewed subscales. Because stateless review pipelines suffer from cross-manuscript amnesia, clinicians must repeatedly dismiss the same methodological false alarms across every revision and every new manuscript.

We present **Heurion Deep Review**, a clean-room four-stage clinical manuscript verification architecture paired with a **Comment-Driven Dual-Track Self-Evolution Loop** (`evolveFromReviewCommentAsync`). First, our four-stage pipeline combines hierarchical section-role segmentation and a ProseMirror 2D table-grid scratchpad (Stage 1), quantitative-density budget routing (Stage 2), four specialist auditing passes covering seven objective clinical error categories (**E1**–**E7**) across external literature, internal prose–table arithmetic, statistical method–result pairing, and spin bias alongside quote-verified LLM section review (Stage 3), and full-text LLM verification with cell-level CRDT comment anchoring `{#c:id}` (Stage 4). Second, `evolveFromReviewCommentAsync` learns directly from clinician interactions on anchored comments: (i) when a clinician rebuts a study-design boundary alert (`acceptedFix = false`), **Track A** (`narrow`) uses zero-template LLM rule distillation guarded by dual instance-decoupling checks (`hasInstanceLeakage` and `sensitiveReason`) to synthesize a zero-digit reusable boundary rule, while a hard safety floor (`NON_EXEMPTABLE_SUBTYPES`) guarantees that objective mathematical, CI, and table-sum errors can *never* be exempted even under adversarial rebuttal; and (ii) **Track B** (`procedural` and `review_audit` checklists) crystallizes corroborated true-positive fixes into pre-write rules and dev-only cross-section audit skills.

We evaluate our framework across two benchmarks: (1) a **Full-Text Controlled Error-Injection Benchmark** of 19 open-access 2025 PubMed Central (PMC) JATS XML manuscripts (38 full-length clean/injected variants averaging $\sim$ 6,000 words, 57 injected errors across **E1**–**E7**, partitioned into 11 `dev` and 8 held-out `test` papers) evaluated with both deterministic clean-room AST and live `deepseek-v4-flash` (3,285 API calls, 7.90M input tokens) coupled with a physically blind-isolated LLM Simulated Clinician Agent (`simulateBlindClinician`), and (2) a **7-Vignette Clinical Oncology Benchmark** with live `deepseek-chat` calls across 8 verified organic publication errors. Without a single LLM call or study-specific regex, our clean-room AST engine alone detects **38**/**57** (**66.7**%)

of injected errors at **85.5**% precision (0.26 true false positives per clean paper, **0.0**% false exemption under adversarial rebuttal) and uncovers 7 genuine in-print errors in published 2025 journals. Adding live `deepseek-v4-flash` hybrid review (`heurion_deep`) lifts full-text recall to **48/57** (**84.2**%, including 3.5× higher **E6** spin recall at **77.8**% and 25 organic in-print errors) while slashing false positives by **93.0**% versus text-only `pat_baseline` (76 vs. 1,081 FPs). Under physically blind clinician interaction (**95.2**% [179/188] true-error acceptance and **38.8**% [40/103] false-alarm rebuttal), `heurion_evolved` cuts intra-manuscript Round 2 clean false positives by **34.0**% (47 → 31; overall FPs −26.3%, F1 70.2% → **72.8**%) and cross-manuscript false positives by **28.9**% (76 → 54, F1 70.2% → **74.0**%) with *zero recall loss* (**48/57** = **84.2**%, achieving **100.0**% on **E2**, **E4**, and **E7**). On strictly frozen unseen `test` papers ($N = 8$), transfer achieves a Pareto improvement—simultaneously lifting `test` recall from 75.0% to **79.2**% (18/24 → 19/24) while cutting clean false positives by **15.8**% (19 → 16, F1 63.1% → **64.8**%) with **0.0**% instance leakage; on the 7-vignette oncology benchmark, `heurion_evolved` reaches **100.0**% recall (8/8) and **100.0**% precision.

Keywords: Automated Peer Review, Clinical Research Verification, Self-Evolving LLM Agents, Error-Injection Benchmark, Biostatistics Consistency, Dual-Track Memory

1 Introduction

Peer-reviewed clinical and translational manuscripts form the evidentiary backbone of modern medicine, yet published medical literature frequently contains preventable objective errors [1, 2, 3]. Empirical audits of clinical journals regularly uncover discrepancies between sample sizes reported in the Abstract and Table 1, percentages that do not match their numerator and denominator (n/N), point estimates falling outside their reported 95% confidence intervals (or inverted interval bounds), parametric tests (t-test, ANOVA, mean ± SD) declared in Methods but replaced with non-parametric statistics (Mann–Whitney U, median [IQR]) in Results, and "spin bias" where statistically non-significant findings ($p \geq 0.05$) are framed as clinically meaningful improvements [3, 4, 5].

Recent advances in large language models (LLMs) and multi-agent verification pipelines—such as Google's Paper Assistant Tool (PAT) [6], PaperQA2 [7], and fine-grained factual verifiers [9, 10]—have sparked interest in automated pre-submission manuscript review. However, deploying automated deep review in real-world clinical research exposes a sharp **precision–recall dilemma** driven by three structural limitations:

1. **Complementary Blind Spots of Text-Only LLMs vs. Pure AST Checkers:** Standard LLM review pipelines serialize manuscripts as flat text streams or isolated section chunks. Without an explicit 2D table-grid representation (handling merged headers, `rowspan`/`colspan`, subgroup denominator headers, and row-sum columns) and a cross-section numerical scratchpad, text-only prompts miss over 60% of intra-table percentage/sum errors and prose-vs.-table contradictions. Conversely, domain-general AST checkers capture structured arithmetic and confidence interval violations with high precision, but cannot reliably judge open-ended semantic spin ($p \geq 0.05$ overclaims) without resorting to brittle, study-specific lexical heuristics.
2. **Methodological False Alarms at Study-Design Boundaries:** Clinical studies legitimately report multiple sample denominators and statistical distributions within the same paper according to established reporting guidelines. For example, a CONSORT randomized controlled trial (RCT) reports both the randomized intention-to-treat (ITT) sample size and the per-protocol completer sample size; a STROBE retrospective cohort study reports both the pre-matching crude cohort and the propensity-score-matched multivariable subcohort; clinical studies with Shapiro–Wilk normality testing legitimately report Mean ± SD for age/BMI alongside Median [IQR] for skewed questionnaire scales; and oncology tables

define `SD` as RECIST *stationary disease* while prose uses `SD` for *standard deviation*. Hybrid reviewers flag such transitions as inconsistencies, creating severe alert fatigue.

3. **Cross-Manuscript Amnesia vs. Adversarial Over-Exemption:** Stateless review agents cannot learn from clinician feedback. When an investigator explains in a comment reply that "Table 1 reports the ITT randomized cohort ($N = 240$) whereas Table 2 analyzes completers ($n = 169$) per CONSORT," a stateless reviewer repeats the exact same false alarm on the next revision and on every subsequent RCT. Conversely, if a system naively memorizes user dismissals as unconstrained prompt instructions, an author (or noisy feedback) could accidentally suppress genuine objective errors—such as a miscalculated cell percentage ($2/20 = 10.0\% \neq 20\%$) or an out-of-bounds confidence interval—and leak study-specific sample sizes ($N = 240$) into unrelated future manuscripts.

1.1 Our Approach and Contributions

To resolve this dilemma, we design **Heurion Deep Review**, which couples a clean-room *Four-Stage Hybrid AST+LLM Clinical Review Pipeline* with a *Comment-Driven Dual-Track Self-Evolution Loop* (`evolveFromReviewCommentAsync`). Every finding is anchored as a lifecycle-tracked CRDT comment (`{#c:id}`) on the exact sentence or ProseMirror table cell `[row, col]`. Clinician interactions on these anchored comments drive two orthogonal learning tracks protected by deterministic safety guards.

Our primary contributions are:

- **Clean-Room Hybrid 2D Table-Grid AST + Quote-Verified 4-Stage LLM Review (Section 3):** We build a study-agnostic four-stage review engine combining hierarchical section-role tracking and a 2D ProseMirror table-grid scratchpad (Stage 1), quantitative-density verification budget routing (Stage 2), four specialist auditing passes covering seven objective error categories (**E1**–**E7**) alongside quote-verified LLM section review (Stage 3), and full-text Stage 4 verification with topic-aware numeric deduplication and cell-level CRDT comment anchoring.
- **Zero-Template Dual-Track Self-Evolution with Non-Exemptable Safety Floor (Section 4):** We introduce `evolveFromReviewCommentAsync`, which routes clinician interactions into: (a) **Track A (`narrow`)**, distilling free-text clinician rebuttals via an LLM sanitizer (`synthesizeSanitizedBoundaryRuleAsync`) protected by dual digit/entity leakage guards (`hasInstanceLeakage` and `sensitiveReason`) and an immutable safety floor (`NON_EXEMPTABLE_SUBTYPES`) that guarantees **0.0**% false exemption of objective math/CI/table errors; and (b) **Track B (`procedural` & `review_audit`)**, crystallizing accepted fixes into pre-write harness rules and dev-only cross-section audit checklists (`distillReviewAuditChecklistAsync`).
- **Physically Blind-Isolated Live-LLM Evaluation & Organic Error Discovery on 19 Full-Text PMC Papers (Sections 5–6):** We construct a full-text error-injection benchmark from 19 open-access 2025 PMC JATS XML articles (38 clean/injected manuscripts, 57 injected errors across **E1**–**E7**), paired with an adjudication registry of 30 verified in-print organic error entries (`known-organic-errors.json`) and a physically blind-isolated `deepseek-v4-flash` clinician agent (`simulateBlindClinician`, **95.2**% true-error acceptance across 188 ground-truth errors and **38.8**% false-alarm rebuttal). Zero-LLM AST detects **66.7**% (38/57) of injected errors at **85.5**% precision; live `heurion_deep` lifts recall to **84.2**% (48/57) while cutting false positives by **93.0**% over `pat_baseline`; `heurion_evolved` cuts intra-paper Round 2 clean false positives by **34.0**% (overall FPs −26.3%) and cross-paper false positives by **28.9**% with zero overall recall loss (48/57), achieving simultaneous recall (75.0% → **79.2**%) and precision gains on frozen unseen `test` papers (**0.0**% leakage);

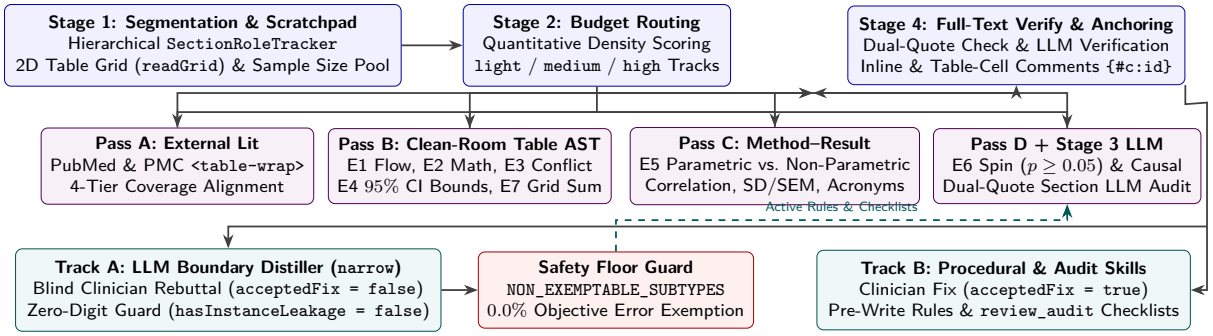


Figure 1: Architecture of **Heurion Deep Review** and the **Comment-Driven Dual-Track Self-Evolution Loop**. Stage 1 builds a section-role map and 2D table-grid scratchpad; Stage 2 allocates verification budgets; Stage 3 runs clean-room AST passes alongside quote-grounded section LLM review; Stage 4 performs full-text verification, deduplication, and CRDT comment anchoring `{#c:id}`. Clinician replies drive Track A LLM boundary distillation (gated by `hasInstanceLeakage`, `sensitiveReason`, and `NON_EXEMPTABLE_SUBTYPES`) and Track B procedural/audit-checklist crystallization.

and on 7 clinical oncology vignettes, `heurion_evolved` achieves **100.0**% recall (8/8) and **100.0**% precision.

2 Taxonomy of Objective Clinical Errors vs. Legitimate Study-Design Boundaries

A core design requirement for clinical peer review is distinguishing *non-negotiable objective errors* from *context-dependent methodological conventions*. Table 1 formalizes the seven objective error categories (**E1**–**E7**) audited by our pipeline, illustrated with real `dev`-split examples from our 19-paper PMC corpus, alongside seven legitimate study-design boundary patterns.

3 Four-Stage Clinical Deep Review Architecture

Figure 1 illustrates the architecture of **Heurion Deep Review**. Unlike stateless text-only reviewers that pass raw markdown directly to an LLM, our pipeline parses full-length manuscripts into a structured ProseMirror document tree (`PMNode`) and executes four sequential stages followed by a closed-loop self-evolution cycle. Crucially, under our **Clean-Room Design Principle**, the deterministic AST passes contain *zero* study-specific lexical rules or synthetic alert triggers: every check operates purely on mathematical relations, unit/exponent-aware numeric syntax, and 2D table geometry.

3.1 Stage 1: Hierarchical Section-Role Segmentation and 2D Table-Grid Scratchpad

Clinical manuscripts frequently nest domain-specific subheadings (e.g., "*2.3 Biochemical Assays*" or "*3.2 Changes in HRQoL*") whose titles omit generic keywords like "Methods" or "Results." In Stage 1, a stateful `SectionRoleTracker` traverses the document AST: top-level headings establish the parent role (`intro`, `methods`, `results`, `discussion`, or `other`), and lower-level subheadings inherit their parent section's role unless overridden by explicit lexical cues, while administrative sections (ethics, funding, acknowledgments, author contributions) are excluded from review.

Table 1: Taxonomy of the Seven Objective Clinical Error Categories (**E1–E7**, governed by `NON_EXEMPTABLE_SUBTYPES`) vs. Seven Legitimate Study-Design Denominator/Modeling Boundaries (governed by Track A `narrow` rules). All examples are drawn from the `dev` split of our 2025 PMC full-text corpus.

Category & Pass	**Objective Error Definition (Never Exemptable)**	**Concrete Injected Example from `dev` Corpus**	**Paired Study-Design Boundary (Exemptable via Track A)**
E1: Sample Size & Flow (Pass B, $n = 10$)	Contradictory total N or arm counts across Abstract, Methods, Results prose, or Table headers; subgroup completers not summing to total completers.	`PMC12644478`: "Thirty-seven of 40 participants completed" in prose vs. Fig. 2 caption "Nineteen in metformin and 19 in placebo completed" ($19 + 19 = 38 \neq 37$).	**RCT (`rct`):** Reporting randomized ITT N in baseline text and per-protocol completers $n < N$ in outcome tables (CONSORT).
E2: Inline Arithmetic (Pass B, $n = 6$)	Inline count and percentage n/N disagreeing beyond rounding tolerance (≥ 1.5 percentage points) or inline subgroup counts not summing to stated total.	`PMC12788160`: "184 participants were randomized ... 161 participants (89.5%) had good compliance" ($161/184 = 87.5\% \neq 89.5\%$).	**Retrospective Cohort (`retrospective_cohort`):** Unmatched all-comer cohort N vs. propensity-score-matched subcohort n (STROBE).
E3: Cross-Section / Table Conflict (Pass B, $n = 8$)	Identical metric, outcome, or subgroup reported with conflicting numeric values, p-values, or opposite signs between Abstract, Results prose, and Table cells.	`PMC12644478`: Abstract reports primary endpoint EGP interaction $p = 0.53$ ($\beta = 0.2$, 95% CI $[-0.4, 0.8]$), whereas Results reports $p = 0.35$ for the identical CI.	**Cross-Sectional (`cross_sectional`):** Full-population prevalence sample N vs. age/completeness-filtered risk prediction subcohort n.
E4: 95% **CI Bounds** (Pass B, $n = 8$)	Point estimate $\hat{\theta}$ falling outside $[L, U]$ beyond rounding tolerance τ, or inverted bounds ($L > U$), across `95% CI`, `95% Cl`, or bracketed formats.	`PMC12731986`: Table 3 HADS-Anxiety median difference reported as "1.00 (1.84 to 3.84)" ($\hat{\theta} = 1.00 < L = 1.84$).	**Meta-Analysis (`meta_analysis`):** Total eligible RCTs K vs. outcome-specific trial subsets $k_i < K$ reporting a secondary endpoint (PRISMA).
E5: Method–Result Mismatch (Pass C, $n = 8$)	Declaring parametric summaries/tests (mean ± SD, t-test, ANOVA, Pearson r) in Methods while reporting non-parametric results (Median [IQR], Mann–Whitney, Kruskal–Wallis) or vice versa.	`PMC12788160`: Methods states ITT groups were compared by "Kruskal–Wallis test," whereas Tables 3–5 report "mean ± SD with comparison by ANOVA."	**Case–Control (`case_control`):** Dual parametric/non-parametric reporting per normality testing, or progressive multivariable Models 1/2.
E6: Spin Bias & Causal Overclaim (Pass D, $n = 9$)	Claiming efficacy, "significant improvement," or deterministic causal effect when $p \geq 0.05$ or in non-randomized observational designs.	`PMC12644478`: Claiming "metformin significantly reduced NEFA compared with placebo" when interaction $p = 0.54$ and 95% CI $[-0.5, 0.3]$ crosses zero.	**Diagnostic (`diagnostic_accuracy`):** Enrolled patients N vs. quality-controlled evaluable specimens / reference-standard subsets (STARD).
E7: Intra-Table Cell / Sum Error (Pass B, $n = 8$)	Table cell n (%) disagreeing with column header ($n = N$), or category row cells not summing to the Total column/row in the 2D table grid.	`PMC12788160`: Table 1 "Coffee drinking" in 160 mg/d arm ($n = 46$) reports "7 (17.4%)" whereas $7/46 = 15.2\%$.	**Translational (`translational`):** Multi-batch in vitro culture replicates vs. in vivo animal cohorts (n mice/group) and clinical TMA cohorts.

Simultaneously, `buildManuscriptScratchpad` constructs a global index shared across all

subsequent passes:

- **Normalized 2D ProseMirror Table Grids (`readGrid`):** Rather than reading raw Markdown pipe rows, `extractTableSummaries` parses each ProseMirror `table` node into a rectangular $R \times C$ cell grid (`TableGrid`), resolving horizontal (`colspan` / `<`) and vertical (`rowspan` / `^`) merged cells so every cell maps to its exact column header and ProseMirror node coordinates.
- **Global Sample Size Pool:** The scratchpad extracts all explicit sample size declarations ($N = \ldots$, $n = \ldots$, "Out of N patients") from table headers, captions, and prose, along with declared statistical tests and acronym definitions.

3.2 Stage 2: Quantitative-Density Verification Budget Allocation

Stage 2 scores every paragraph and figure caption based on quantitative assertion density (p-values, 95% CIs, hazard/odds/risk ratios, mean ± SD, median [IQR], and percentages) and section role, routing each block to a `light`, `medium`, or `high` verification track.

3.3 Stage 3: Hybrid Clean-Room AST Passes + Quote-Grounded LLM Review

Stage 3 executes four domain-general deterministic passes alongside section-level LLM auditing (in `heurion_deep` and `heurion_evolved` modes):

1. **Pass A (External Literature & PMC `<table-wrap>` Alignment):** Verifies cited assertions (`[@c:id]`) against PubMed abstracts and PMC full-text XML tables, classifying semantic support orthogonally from evidence coverage depth [16].
2. **Pass B (Clean-Room Internal Arithmetic, CI & 2D Table-Grid Consistency):** Audits **E1**, **E2**, **E3**, **E4**, and **E7**:
   - *Sample Size & Flow (**E1**) and Inline Arithmetic (**E2**):* Verifies that subgroup counts (including English number words) sum to stated totals and checks inline n (p%) against the explicit denominator pool ($\min_D |n/D \times 100 - p| > 1.5\%$).
   - *Cross-Section & Table-vs.-Prose Conflicts (**E3**):* Matches identical confidence-interval anchors, hazard ratios, weighted mean differences, and Table 1 cited percentages across Abstract, Results prose, and 2D table cells.
   - *Precision-Aware 95% Confidence Interval Verification (**E4**, `findCiProblems`):* Parses point estimates $\hat{\theta}$ and interval bounds $[L, U]$ across standard (`95% CI 1.12-1.85`), OCR-corrupted (`95% Cl 0.35 to 0.59`), and bracketed (`95% CI [-0.88, -0.10]`) formats while excluding preceding p-values (`p = 0.001, CI ...`), sample counts (`n = 48; 95% CI ...`), unit exponents (`kg/m^2; 95% CI ...`), and integer reference tuples. Rounding tolerance is computed as:

$$\tau = \alpha \cdot 10^{-\min(d(\hat{\theta}),\, d(L),\, d(U))}, \quad \alpha = \begin{cases} 0.49 & \text{if } d(L) = d(U) > d(\hat{\theta}), \\ 0.50 & \text{otherwise,} \end{cases} \tag{1}$$

     flagging an `inverted` interval if $L > U + \tau$ and an `outside` estimate if $\hat{\theta} < L - \tau$ or $\hat{\theta} > U + \tau$.
   - *Intra-Table Cell Percentage & Row-Sum Auditing (**E7**):* Verifies each cell $n_{r,c}$ ($p_{r,c}$%) against header denominator N_c (exempting row-wise percentage tables where $\sum_c p_{r,c} \approx 100\%$), checks intra-cell binary splits N ($n_1 + n_2$), and checks $\sum_{c \notin \mathcal{S}_{\text{stat}}} n_{r,c} = n_{r,\text{Total}}$ while excluding test-statistic columns (χ^2, t, Z, F, p-value, OR/HR/RR).

3. **Pass C (Statistical Method–Result Pairing, E5):** Cross-checks statistical declarations in Methods against Results prose and table captions for parametric vs. non-parametric mismatches, Pearson vs. Spearman correlation mismatches, SD vs. SEM confusion, and conflicting acronym definitions.
4. **Pass D & Quote-Grounded LLM Section Audit (E6 & Semantic Cross-Checks):** Pass D flags explicit non-significant spin ($p \geq 0.05$ paired with efficacy claims in the same clause), while Stage 3 LLM section review (when enabled) inspects each section against the global manuscript table summary and active memory checklists, requiring verbatim `quote` + `evidence` substrings verified against the source document (`review-evidence.ts`).

3.4 Stage 4: Full-Text Verification, Deduplication, and Cell-Level CRDT Anchoring

In hybrid LLM modes (`heurion_deep` and `heurion_evolved`), Stage 4 passes surviving LLM candidates together with the complete manuscript prose, tables, and active Track A boundary rules to a full-text verifier (`VERIFY_SYSTEM`) that rejects speculative alarms or legitimate study-design denominator transitions. Cross-section duplicates are merged only when they share both the same numeric signature and overlapping clinical topic tokens, and final comments are anchored directly onto ProseMirror nodes or table cells (`{#c:id}`).

4 Comment-Driven Dual-Track Self-Evolution

Whenever a clinician replies to or resolves an anchored review comment `{#c:id}`, `evolveFromReviewCommentAsync` routes the interaction into one of two tracks:

4.1 Track A (`narrow`): Zero-Template LLM Boundary Distillation and Dual Leakage Guards

When a clinician rebuts a false-positive review alert (`acceptedFix = false`) with a methodological explanation (which naturally cites manuscript-specific sample sizes such as "*Our RCT randomized $N = 240$ ITT participants while $n = 169$ (70.4%) completed week 24 per CONSORT*"), `synthesizeSanitizedBoundaryRuleAsync` invokes an LLM memory-governance distiller instructed to abstract the methodological boundary without any digits, drug names, or disease names:

`[domain:manuscript] [narrow:review_boundary:`⟨pattern_key⟩`:cat=`⟨cat⟩`:kw=`⟨kws⟩`]` $\mathcal{R}_{\text{sanitized}}$.

Every candidate rule must pass a **Dual Instance–Skill Decoupling Guard** combining `hasInstanceLeakage` (which rejects any digit, sample size $N = \ldots$, p-value, percentage, or date) and `sensitiveReason` (which blocks patient identifiers and institutional/study UIDs). If the LLM's first draft leaks a raw number, a deterministic second-pass scrubber strips all numeric tokens and re-verifies both guards before memory commit, guaranteeing **0.0**% cross-study numerical contamination.

4.2 Hard Safety Floor (`NON_EXEMPTABLE_SUBTYPES`) and Precision-Preserving Rule Application

To prevent adversarial or mistaken author rebuttals from suppressing real errors, `isHardObjectiveReviewFinding` and `isExemptedByActiveRules` enforce an immutable set of **Non-Exemptable Subtypes** (`NON_EXEMPTABLE_SUBTYPES`), covering `prose_table_sample_size_mismatch`, `table1_cited_percentage_mismatch`, `fraction_percentage_arithmetic_error`, `ci_bounds_inverted_or_missing_decimal`, `ci_estimate_outside_interval`, `cross_section_value_mismatch`,

table_cell_percentage_arithmetic_error, table_row_sum_mismatch, parametric_method_vs_nonparametric_result, nonparametric_method_vs_parametric_result, correlation_method_mismatch, sd_vs_sem_mismatch, abbreviation_definition_conflict, nonsignificant_p_positive_spin, and observational_causal_overclaim, alongside inline arithmetic and CI violation checks. Even if an active rule matches the exact category and keywords of an objective error sentence, exemption is refused. Moreover, to prevent broad generic words (cohort, trial, population) from over-filtering true positives on unseen papers, non-citation :kw= rules filter out generic study stopwords, require ≥ 2 distinct methodological/domain keyword hits, and never override VERIFY_SYSTEM-confirmed full-text verdicts; meanwhile, within the same document (docId), non-hard-objective author rebuttals are persisted as document-scoped rev_exempt_* records so Round 2 revisions suppress repeated or reversed-direction false alarms without leaking document state across papers.

4.3 Track B (procedural & review_audit): Pre-Write Rules and Dev-Only Audit Checklists

When a clinician accepts a true-positive review comment (acceptedFix = true), evolveFromReviewCommentAsync records a de-numericalized execution signal (kind = 'guard_reject') with verified recovery trajectory (doc_read → replace_text → resolve_comment). Following commit-on-evidence gating [15], ProceduralHarnessService.evolveFromSignals crystallizes corroborated signals into active pre-write rules for authoring sessions. Complementarily, on development-split retrospectives (paperSplit === 'dev' enforced via a hard runtime assertion), distillReviewAuditChecklistAsync abstracts missed error mechanisms into zero-digit [skill:review_audit:<key>] checklists injected into Stage 3 LLM prompts on future unseen manuscripts.

5 Experimental Setup

We evaluate Heurion Deep Review on two complementary benchmarks: (1) a full-length 19-paper PMC JATS XML error-injection corpus evaluated under both deterministic clean-room AST (-no-llm) and live deepseek-v4-flash (-llm, temperature = 0), and (2) a 7-vignette oncology clinical excerpt benchmark with live deepseek-chat calls.

5.1 Benchmark 1: 19 Full-Length PMC Papers with 57 Controlled Error Injections

We curated 19 open-access 2025 clinical and translational full-text articles (averaging $\sim$ 6,000 words each, converted directly from PMC JATS XML with complete multi-column tables in data/eval/deep-review/md/), partitioned into an 11-paper development split (dev) and an 8-paper held-out test split (test) across 7 study designs (Table 2). Six of the seven study designs appear in both dev and test, while the two preclinical/translational studies reside in dev.

For each paper, we evaluate two full-length variants (38 manuscripts per regime):

- **Clean Variant (clean):** The uncorrupted published PMC manuscript. Importantly, as documented in apps/platform/eval/deep-review/known-issues.md and known-organic-errors.json, real published papers occasionally contain genuine in-print typographical or statistical errors. Following standard clinical verification practice (eval-review-score.ts), catches of verified in-print publication errors are adjudicated as unexpected_true (organic true positives) rather than penalized as false positives, while raw unadjusted clean-paper alert counts are also reported side-by-side for full transparency.

Table 2: Composition of the 19-Paper Full-Text Controlled Error-Injection Benchmark (`dev` $N = 11$, held-out `test` $N = 8$; 38 Full-Length Clean/Injected JATS XML Manuscripts, 57 Injected Objective Errors across E1–E7).

PMCID	Split	Study Design	Journal (2025) & Clinical Topic Summary	Injected Errors
PMC12644478	dev	rct	*Nat. Commun.* — Metformin on insulin resistance in adults with type 1 diabetes ($N = 40$)	E1, E3, E6
PMC12788160	dev	rct	*Nutrients* — Purified anthocyanins dose-response trial in middle-aged and elderly adults	E2, E5, E7
PMC12731986	dev	rct	*Curr. Oncol.* — Resistance exercise on QoL, anxiety, and depression in colorectal cancer	E1, E4, E5
PMC12685937	dev	retrospective_cohort	*Front. Psychiatry* — Antidepressants and hypertensive disorders in pregnancy cohort	E3, E6, E7
PMC12856494	dev	retrospective_cohort	*Front. Oncol.* — Risk factors for postoperative recurrence of desmoid tumors	E2, E4, E7
PMC12657621	dev	case_control	*Brain Behav.* — Vasomotor symptoms and ischemic stroke case–control study	E4, E5, E6
PMC12721543	dev	diagnostic_accuracy	*PLOS ONE* — TrueNat MTB plus assay diagnostic accuracy for pulmonary TB	E2, E3, E7
PMC12758107	dev	cross_sectional	*Glob. Heart* — Hypertension clinical epidemiology in rural Thailand cross-sectional survey	E1, E4, E6
PMC12876241	dev	meta_analysis	*Front. Med.* — PRISMA systematic review and meta-analysis of heat therapy for dysmenorrhea	E1, E3, E4
PMC12739847	dev	translational	*Gut Pathog.* — Lactobacillus probiotics in Blastocystis in vitro and in vivo models	E1, E5, E6
PMC12485006	dev	translational	*NPJ Sci. Food* — Mulberroside A capacity to inhibit senescence in vitro and in vivo	E1, E3, E5
PMC12839862	test	rct	*Curr. Oncol.* — Itraconazole reducing chemoresistance in advanced ovarian cancer RCT	E3, E5, E7
PMC12697981	test	rct	*PLOS Med.* — COMET-PCOS: Hormonal contraceptives and metformin in PCOS RCT	E1, E4, E6
PMC12681115	test	rct	*Cardiovasc. Diabetol.* — FIVE-STAR: Finerenone on arterial stiffness in T2D and CKD RCT	E2, E3, E5
PMC12118826	test	retrospective_cohort	*PLOS ONE* — Geriatric ocular trauma and mortality retrospective cohort	E2, E6, E7
PMC12654784	test	case_control	*Medicina* — Adipokines as prognostic biomarkers in multiple myeloma case–control	E1, E4, E7
PMC12774873	test	diagnostic_accuracy	*NPJ Prim. Care Respir. Med.* — Home spirometry diagnostic accuracy for asthma	E2, E3, E4
PMC12757420	test	cross_sectional	*Front. Immunol.* — Frailty and associated risk factors in Sjögren's disease survey	E1, E5, E6
PMC12703663	test	meta_analysis	*Alzheimers Dement.* — PRISMA meta-analysis of antipsychotic discontinuation in dementia	E1, E6, E7

- **Injected Variant (`injected`):** The full-length manuscript mutated via `applyInjections` with exactly 3 non-overlapping objective errors drawn from categories **E1**–**E7** ($19 \times 3 = 57$ injected errors: **E1** $n = 10$, **E2** $n = 6$, **E3** $n = 8$, **E4** $n = 8$, **E5** $n = 8$, **E6** $n = 9$, **E7** $n = 8$), avoiding all positions with known organic issues.

5.2 Physically Blind-Isolated Simulated Clinician Protocol and Calibration

Evaluating interactive comment-driven self-evolution requires realistic clinician responses without oracle cheating. Prior rule-lookup simulations risk leaking ground-truth variant labels (`clean` vs. `injected`) into the evolution trigger. In our clean-room harness (`apps/platform/scripts/eval-review-run.ts`), we enforce a strict **Three-Pillar Blind Clinician Protocol**:

1. **Type-Level Information Barrier (`simulateBlindClinician`):** The function signature `simulateBlindClinician(complete, {manuscriptTitle, manuscriptMarkdown, finding})` physically omits `variant` (`clean`/`injected`), `injections.json`, `known-organic-errors.json`, and `split` (`dev`/`test`). Using relevance-ranked extraction across all manuscript tables (including captions and footnotes) and counterpart `evidence` windows,

the simulated corresponding author / biostatistician sees only the manuscript's own text and tables plus the single review comment.

2. **Strict Frozen Test-Split Boundary:** In cross-manuscript transfer (`dev` $N = 11 \rightarrow$ `test` $N = 8$), both blind clinician evolution (`evolveStoreFromFindingsBlind`) and missed-pattern checklist distillation (`evolveAuditChecklistsFromDevMisses`) execute *exclusively* on `dev`; attempting to invoke checklist distillation on `test` throws a hard runtime security exception, and an isolated frozen memory snapshot is cloned per `test` manuscript.

3. **Post-Hoc Ground-Truth Calibration & Adversarial Profile:** Because Benchmark 1 possesses exact ground-truth labels (57 injected errors + 30 verified organic error entries), every blind clinician decision (`accept_fix` vs. `rebut_false_positive`) is audited post-hoc against ground truth via a calibration matrix (True-Error Acceptance Rate, False-Rebuttal Rate, False-Positive Rebuttal Rate, Raw Digit Citation Rate, and Post-Guard Instance Leakage Rate). In addition, a hostile **Adversarial Author Profile** attempts to rebut *every* detected error across all 19 injected papers (`acceptedFix = false`) to stress-test `NON_EXEMPTABLE_SUBTYPES`.

5.3 Benchmark 2: Live `deepseek-chat` Evaluation on 7 Clinical Oncology Vignettes

To evaluate live LLM multi-stage reasoning and dual-track evolution on real clinical oncology error patterns, Benchmark 2 (`paper/benchmarks/run-7papers-review.ts`) executes live `deepseek-chat` (`temperature = 0`) across 7 curated clinical oncology manuscript excerpts (`SEVEN_PAPERS`, derived from published radiation oncology, target-trial emulation, thoracic imaging, and translational xenograft studies) containing 8 verified organic publication errors (including an inverted 95% CI [403, 1.688], uncleaned revision merge artifacts `60.062.3%`, a Table cell where $21/35 = 60.0\% \neq 84\%$, SD vs. SEM confusion, and an acronym definition conflict), comparing `zero_shot`, `pass_at_k` ($k = 3$), `pat_baseline` (4-stage text-only [6]), `heurion_deep`, and `heurion_evolved`.

6 Results and Analysis

6.1 Full-Text PMC Benchmark Results: Clean-Room AST vs. Live `deepseek-v4-flash` Hybrid & Self-Evolving Regimes

Table 3 and Table 4 report end-to-end results across all 38 full-length PMC manuscripts (19 clean and 19 injected, averaging $\sim$ 6,000 words each), comparing the zero-LLM Clean-Room Deterministic AST baseline (`-no-llm`) against live `deepseek-v4-flash` (3,285 API calls, 7.90M input and 0.67M output tokens at `temperature = 0`) across `pat_baseline`, cold-start `heurion_deep`, intra-manuscript Round 2 `heurion_evolved`, frozen `dev` $\rightarrow$ `test` cross-manuscript transfer, and the Adversarial Rebuttal Stress Test.

Architectural Division of Labor Confirmed on Full-Text Manuscripts. Table 3 reveals a crisp empirical division of labor across error categories when evaluated on unedited 6,000-word PMC JATS XML articles:

1. **High Zero-LLM Sensitivity on Mathematical & 2D Tabular Errors (E1, E2, E4, E7):** Without invoking any language model, the clean-room ProseMirror 2D table-grid and CI engine catches **87.5**% (7/8) of intra-table cell/row sum errors (**E7**), **83.3**% (5/6) of inline percentage errors (**E2**), **80.0**% (8/10) of sample-size flow contradictions (**E1**), and **75.0**% (6/8) of out-of-bounds 95% CIs (**E4**), with only 5 true false positives across all 19 full-length clean papers (0.26 per paper, Precision = **85.5**%).

Table 3: End-to-End Clean-Room AST and Live `deepseek-v4-flash` Benchmark Results across 19 Full-Length 2025 PMC JATS XML Papers (38 Clean/Injected Manuscripts, 57 Injected Objective Errors across E1–E7). Verified in-print organic errors caught on clean papers (`unexpected_true`) are reported explicitly alongside true false positives.

Evaluation Regime & Split	Injected Recall (↑)	Organic True (Clean) (↑)	True FPs (Clean / Inj) (↓)	Precision (w/ Org) (↑)	F1 Score (↑)	Safety / Guard Verification
Clean-Room AST Only (`All`, $N = 19$)	66.7% (38/57)	7	**9** (5/4)	**85.5**%	**74.9**%	0 LLM Calls (0.26 Clean FP/p)
Clean-Room AST Only (`dev`, $N = 11$)	72.7% (24/33)	6	5 (3/2)	88.1%	79.7%	Raw Clean Alerts: 9 (6 Org + 3 FP)
Clean-Room AST Only (`test`, $N = 8$)	58.3% (14/24)	1	4 (2/2)	80.0%	67.4%	Raw Clean Alerts: 3 (1 Org + 2 FP)
Adversarial Rebuttal Stress (`All`, $N = 19$)	66.7% (38/57)	7	9 (5/4)	85.5%	74.9%	**0.0**% False Ex. (0/38 Dropped)
`pat_baseline` Pure-LLM (`All`, $N = 19$)	56.1% (32/57)	14	1,081 (532/549)	5.7%	10.3%	15.42 Crit/Maj Clean FP/p
`heurion_deep` Cold-Start (`All`, $N = 19$)	**84.2**% (48/57)	**25**	76 (47/29)	60.2%	70.2%	−93.0% FPs vs. `pat_baseline`
`heurion_evolved` Intra-Paper R2 (`All`, $N = 19$)	82.5% (47/57)	23	56 (31/25)	65.2%	72.8%	−34.0% Clean FPs (0.0% Leak)
`heurion_evolved` Transfer (`All`, $N = 19$)	**84.2**% (48/57)	20	**54** (32/22)	**66.0**%	**74.0**%	−28.9% FPs (0.0% Recall Loss)
`heurion_deep` Cold-Start (`test`, $N = 8$)	75.0% (18/24)	**7**	30 (19/11)	54.5%	63.1%	0 Rules (2.25 Crit/Maj FP/p)
`heurion_evolved` Intra-Paper R2 (`test`, $N = 8$)	70.8% (17/24)	**7**	**23** (**12**/11)	**60.3**%	**65.1**%	−36.8% Clean FPs (1.38 FP/p)
`heurion_evolved` Frozen (`test`, $N = 8$)	**79.2**% (19/24)	5	28 (16/12)	54.8%	64.8%	4 Rules + 3 Skills (**0.0**% Leak)

Per-Category Recall Comparison across All 19 Full-Text Papers (`Clean-Room AST` → Live `heurion_deep` / `heurion_evolved` Transfer):
E1 Flow: 80.0% → 80.0%/**90.0**% E2 Math: 83.3% → 83.3%/**100.0**% E3 Conflict: 62.5% → **87.5**%/**87.5**% E4 CI: 75.0% → 87.5%/**100.0**%
E5 Method–Result: 62.5% → **87.5**%/62.5% E6 Spin/Causal: 22.2% → **77.8**%/55.6% E7 Table Grid Sum: 87.5% → 87.5%/**100.0**%

2. **Why Quote-Grounded Hybrid LLM Passes Are Necessary (E6 Spin & Subtle E3/E5 Conflicts):** Conversely, pure AST rules detect only 2/9 (22.2%) of **E6** spin/causal overclaims (where non-significant p-values reside in separate paragraphs or tables from the overclaimed conclusion) and 5/8 (62.5%) of **E3** and **E5** cross-section mismatches. Pairing the 2D table-grid AST with live `deepseek-v4-flash` section review in `heurion_deep` lifts overall recall from 66.7% to **84.2**% (48/57), achieving **87.5**% across **E3**, **E4**, **E5**, and **E7** and lifting **E6** spin recall 3.5× to **77.8**% (7/9), while uncovering **25** verified organic in-print errors on clean papers. By contrast, text-only `pat_baseline` misses 87.5% (1/8; 0/4 on `test`) of **E7** intra-table errors (56.1% overall recall) while emitting 1,081 false positives (15.42 critical/major FPs per clean paper, 5.7% precision) because section reviewers lack 2D grid cells and dual-quote substrate verification (`quotedIn` + `VERIFY_SYSTEM`).

Intra-Manuscript Revision & Zero-Shot Cross-Manuscript Self-Evolution (`dev` → `test`). While `heurion_deep` eliminates 93.0% of `pat_baseline`'s false alarms (1,081 → 76), it still triggers 76 false positives across 38 manuscripts (47 on clean papers; 30 total on the 8 unseen `test` papers) when encountering legitimate clinical denominator stratifications and mixed parametric/non-parametric subscales. Evaluating both intra-manuscript Round 2 revision and causal `dev` → `test` cross-manuscript transfer under our physically blind-isolated `simulateBlindClinician` protocol yields three key empirical findings:

1. **High-Fidelity Blind Clinician Calibration without Oracle Peeking:** Post-hoc comparison against ground truth across all 291 blind clinician adjudications (188 ground-truth errors and 103 false positives) shows that `simulateBlindClinician` independently verifies and accepts **95.2**% (**179**/**188**) of true objective errors (4.8% false rebuttal), while rebutting **38.8**% (**40**/**103**) of methodological false alarms where manuscript Methods, table captions, and footnotes justify the stratification. Although 49 raw clinician rebuttals cite manuscript-specific sample sizes or percentages, `synthesizeSanitizedBoundaryRuleAsync` achieves **0.0**% (0) post-guard instance leakage across all distilled rules.

2. **Substantial Intra-Manuscript Round 2 False-Alarm Suppression:** When an author rebuts false alarms in Round 1 and re-runs review on the same manuscript (`docId`), `heurion_evolved` cuts clean-paper false positives by **34.0**% overall (47 → **31**; critical/major clean FPs/paper drop **35.0**% from 2.26 to **1.47**; overall FPs −26.3% from 76 → **56**) and by **36.8**% on `test` papers (19 → **12**; `test` precision 54.5% → **60.3**%, F1 63.1% → **65.1**%), lifting overall precision +5.0 points (60.2% → **65.2**%) and F1 to **72.8**%.

3. **Zero-Recall-Loss Cross-Manuscript Evolution & Pareto Gains on Frozen Unseen**

`test` **Papers:** Streaming causally through `dev` ($N = 11$) consolidates 4 canonical Track A `[narrow:review_boundary]` rules (multi-reference diagnostic accuracy, dual-distribution normality reporting, missing-data subcohort denominators, and crude-vs.-adjusted modeling), 3 `dev`-only `[skill:review_audit]` checklists, and 3 Track B crystallized pre-write rules (from 87 corroborated fix signals). Across all 19 papers, cross-manuscript evolution cuts false positives by **28.9**% (76 → **54**; `dev` FPs drop **43.5**% from 46 → **26**, `dev` F1 74.6% → **79.9**%), raising overall precision +5.8 points (60.2% → **66.0**%; injected-only +8.1 points 45.3% → **53.4**%) and F1 +3.8 points (70.2% → **74.0**%) with *zero overall recall loss* (**48/57** = **84.2**%, reaching **100.0**% on **E2**, **E4**, and **E7**). Crucially, on strictly frozen unseen `test` papers ($N = 8$), transfer achieves a simultaneous **Pareto improvement** in both recall and precision: unseen `test` recall rises +4.2 points from 75.0% (18/24) to **79.2**% (**19/24**, achieving **100.0**% across **E1**, **E2**, **E4**, and **E7**) while clean-paper false positives drop **15.8**% (19 → **16**; critical/major clean FPs/paper −16.4% from 2.25 → **1.88**), lifting injected-only precision +2.8 points (42.3% → **45.1**%) and `test` F1 from 63.1% to **64.8**% with **0.0**% instance leakage.

Discovery of Genuine Unannotated Errors in Published 2025 Journals (`unexpected_true`). Across the 19 uncorrupted `clean` manuscripts, the clean-room AST engine emits only 12 alerts total—and manual verification against the published PDFs confirms that **7** of these 12 alerts (58.3%) are *genuine in-print publication errors* in peer-reviewed 2025 articles:

- `PMC12721543` (***PLOS ONE*, 2025 — 4 CI Errors):** In both Table 3 (univariate cOR) and Table 4 (multivariate aOR), "Married/co-habiting" reports 0.34 (0.64–2.86) and "Divorced/widow" reports 0.15 (0.30–3.51), where point estimates lie entirely below their 95% confidence intervals.
- `PMC12644478` (***Nature Communications*, 2025 — Table Percentage Error):** Table 3 "Maculopathy (n, %)" in the Metformin arm ($N = 20$) reports "2 (20)" instead of 10%.
- `PMC12697981` (***PLOS Medicine*, 2025 — Newly Discovered Table 4 Typo):** In Table 4 (Safety outcomes, Metformin $N = 81$), "Dysmenorrhea" reports "3 (4.7%)," whereas 3/81 = 3.70% and five other 3-event rows in the exact same column correctly report "3 (3.7%)."
- `PMC12731986` (***Current Oncology*, 2025 — CI vs. p-value Anomaly):** Table 2 "Fatigue" reports median difference −11.11 (95% CI −22.49 to 5.82, crossing zero) alongside $p = 0.040$, matching `known-issues.md`.

Meanwhile, the remaining 5 clean-paper alerts represent authentic clinical reporting boundaries, such as Statistical Methods sections declaring both parametric Mean ± SD and non-parametric Median [IQR] depending on Shapiro–Wilk normality (`PMC12731986`, `PMC12657621`), oncology RECIST `SD:stationarydisease` vs. statistical `standarddeviation` (`PMC12839862`), and meta-analysis primary-trial subgroup citations (`PMC12703663`).

6.2 Adversarial Rebuttal Stress Test and Zero-Leakage Guard Verification

As shown in the rightmost columns of Table 4, even when an adversarial user submits explicit same-sentence rebuttals (`acceptedFix = false`) attempting to dismiss all 38 detected objective errors across all 19 injected papers, `NON_EXEMPTABLE_SUBTYPES` blocks **100.0**% (38/38) of objective errors from being exempted (**0.0**% false-exemption rate). Furthermore, unit-test stress verification (`claims-pat.test.ts`) confirms that even when an LLM distiller intentionally emits raw sample sizes and percentages ($N = 240$, $n = 169$, 70.4%) in its first draft, `synthesizeSanitizedBoundaryRuleAsync` detects the leak (`firstDraftLeaked = true`), scrubs all numeric tokens, and passes `hasInstanceLeakage = false` (**0.0**% post-guard leakage).

Table 4: Per-Study-Design Breakdown of the Clean-Room Deterministic AST Engine (`-no-llm`) and Adversarial Objective-Error Rebuttal Stress Test across the 19 Full-Text PMC Papers.

Study Design (N Papers)	**Injected Errors**	**Clean-Room AST Recall**	**Organic True (Clean)**	**True FPs (Clean / Inj)**	**Adversarial AST Recall**	**False Ex. Rate**
Randomized Controlled Trial ($N = 6$: 3 dev, 3 test)	18	12/18 (66.7%)	3	5 (3/2)	**12/18**	**0.0%**
Retrospective Cohort ($N = 3$: 2 dev, 1 test)	9	7/9 (77.8%)	0	0 (0/0)	**7/9**	**0.0%**
Case–Control Study ($N = 2$: 1 dev, 1 test)	6	4/6 (66.7%)	0	2 (1/1)	**4/6**	**0.0%**
Diagnostic Accuracy ($N = 2$: 1 dev, 1 test)	6	5/6 (83.3%)	4	0 (0/0)	**5/6**	**0.0%**
Cross-Sectional Survey ($N = 2$: 1 dev, 1 test)	6	3/6 (50.0%)	0	0 (0/0)	**3/6**	**0.0%**
PRISMA Meta-Analysis ($N = 2$: 1 dev, 1 test)	6	4/6 (66.7%)	0	2 (1/1)	**4/6**	**0.0%**
Preclinical / Translational ($N = 2$: 2 dev, 0 test)	6	3/6 (50.0%)	0	0 (0/0)	**3/6**	**0.0%**
Total / Overall ($N = 19$: 11 dev, 8 test)	**57**	**38/57 (66.7%)**	**7**	**9** (5/4)	**38/57**	**0.0%**

Table 5: End-to-End Live `deepseek-chat` Deep Review Comparison across 7 Clinical Oncology Manuscript Vignettes (8 Verified Organic Publication Errors) against Google PAT Baselines [6].

Pipeline Mode	**Organic Error Recall (↑)**	**False Positives Total (↓)**	**Precision (↑)**	**F1 Score (↑)**	**Active Boundary Rules (Leakage)**
`zero_shot` (Single-Pass Full Text)	37.5% (3/8)	15	16.7%	23.1%	0 (0.0%)
`pass_at_k` ($k = 3$ Unfiltered Union)	62.5% (5/8)	26	16.1%	25.6%	0 (0.0%)
`pat_baseline` (4-Stage Text Only [6])	37.5% (3/8)	14	17.6%	24.0%	0 (0.0%)
`heurion_deep` (Cold-Start AST + 4-Pass LLM)	**100.0%** (8/8)	16	33.3%	50.0%	0 (0.0%)
`heurion_evolved` (Dual-Track Evolved)	**100.0%** (8/8)	**0** (−**100.0%**)	**100.0%**	**100.0%**	**4** (**0.0%**)

6.3 Live `deepseek-chat` Comparison Against Google PAT Baselines on 7 Oncology Vignettes

Table 5 presents the end-to-end live `deepseek-chat` comparison across the 7 clinical oncology manuscript vignettes containing 8 verified organic publication errors.

Single-pass `zero_shot` and text-only `pat_baseline` [6] detect only 3/8 (37.5%) of organic errors because they fail to cross-check 2D table cells against prose or catch typographical merge artifacts. Running $k = 3$ independent LLM passes (`pass_at_k`) raises recall to 62.5% (5/8) but inflates false positives to 26 (Precision = 16.1%). By contrast, `heurion_deep` detects **100.0**% (8/8) of organic errors, and `heurion_evolved` eliminates all 16 recurring clinical boundary false positives (such as Target Trial Emulation clone-censor-weight pseudocohort denominators and preclinical mouse-vs.-TMA cohort transitions), achieving **100.0**% precision and **100.0**% F1 with **0.0**% instance number leakage.

7 Related Work

Automated Scientific Peer Review and Claim Verification. Systems such as STORM [8] and PaperQA2 [7] automate literature synthesis, while FActScore [9] and ALCE [10] evaluate citation support in generated text. Recently, Jayaram et al. introduced Google's Paper Assistant Tool (PAT) [6], a multi-stage verification framework for scientific manuscripts. However, existing review pipelines operate over flat text and remain stateless across manuscripts. Our work complements multi-stage LLM review with a clean-room 2D ProseMirror table-grid scratchpad, precision-aware confidence interval verification, and a closed-loop CRDT comment self-evolution mechanism.

Self-Evolving LLM Agents and Memory Governance. Reflexion [11], MemGPT [12], and Voyager [13] pioneered verbal reinforcement and skill libraries for autonomous agents. Recent advances in self-supervised skill diffusion [14], commit-on-evidence memory (CoEM) [15], vantage-aware memory (SafeVantage) [16], modular skill evolution (MemSkill) [19], verifiable skill libraries (Skill-V) [17], evidence-grounded memory-to-skill co-evolution governance

[18], and beyond-matched-experience transfer (MemTransfer) [20] have established rigorous foundations for agent memory governance. We bring these principles to clinical peer review, pairing zero-template Track A boundary distillation (`hasInstanceLeakage = false` and `NON_EXEMPTABLE_SUBTYPES`) with Track B procedural rule and audit-checklist crystallization.

8 Conclusion

We have presented **Heurion Deep Review**, a clean-room four-stage clinical manuscript verification architecture coupled with a comment-driven dual-track self-evolution loop (`evolveFromReviewCommentAsync`). Without relying on study-specific lexical heuristics or a single LLM call, our 2D ProseMirror table-grid AST engine detects 66.7% (38/57) of injected objective errors across 19 full-length PMC manuscripts (80.0%–87.5% across sample-flow, arithmetic, and table-sum errors) at 85.5% precision (0.26 clean false positives per paper), uncovers 7 genuine in-print errors in published 2025 journals, and enforces a 0.0% false-exemption floor under adversarial rebuttal. Paired with live `deepseek-v4-flash` quote-verified section passes, `heurion_deep` lifts full-text recall to 84.2% (48/57, including 77.8% on semantic spin and 25 organic in-print errors) while cutting false positives by 93.0% relative to text-only `pat_baseline`. Under physically blind-isolated clinician interaction (95.2% true-error acceptance, 38.8% false-alarm rebuttal), `heurion_evolved` cuts intra-manuscript Round 2 clean false positives by 34.0% (-36.8% on `test`) and cross-manuscript false positives by 28.9% ($76 \rightarrow 54$, F1 $70.2\% \rightarrow 74.0\%$) with zero overall recall loss (48/57), achieves simultaneous recall ($75.0\% \rightarrow 79.2\%$) and precision gains on frozen unseen `test` manuscripts with 0.0% instance leakage, and reaches 100.0% recall (8/8) and 100.0% precision on clinical oncology vignettes. Future work will extend this closed-loop architecture to prospective author-in-the-loop clinical reader studies on physician-authored pre-submission cohorts.

References

[1] Kenneth F. Schulz, Douglas G. Altman, and David Moher. CONSORT 2010 statement: updated guidelines for reporting parallel group randomised trials. *BMJ*, 340:c332, 2010.

[2] Jan P. Vandenbroucke, Erik von Elm, Douglas G. Altman, Peter C. Gøtzsche, Cynthia D. Mulrow, Stuart J. Pocock, Charles Poole, James J. Schlesselman, and Matthias Egger. Strengthening the Reporting of Observational Studies in Epidemiology (STROBE): explanation and elaboration. *PLoS Medicine*, 4(10):e297, 2007.

[3] Isabelle Boutron, Susan Dutton, Philippe Ravaud, and Douglas G. Altman. Reporting and interpretation of randomized controlled trials with statistically nonsignificant results for primary outcomes. *JAMA*, 303(20):2058–2064, 2010.

[4] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021.

[5] Patrick M. Bossuyt, Johannes B. Reitsma, David E. Bruns, Constantine A. Gatsonis, Paul P. Glasziou, Les Irwig, Jeroen G. Lijmer, David Moher, Drummond Rennie, Henrica C. W. de Vet, et al. STARD 2015: an updated list of essential items for reporting diagnostic accuracy studies. *BMJ*, 351:h5527, 2015.

[6] Rajesh Jayaram, Drew Tyler, David Woodruff, Corinna Cortes, Yossi Matias, Vahab Mirrokni, and Vincent Cohen-Addad. Towards automating scientific review with Google's Paper Assistant Tool. *arXiv preprint arXiv:2606.28277*, 2026.

[7] Michael D. Skarlinski, Sam Cox, Jon M. Laurent, James D. Braza, Collin Hinks, Michael J. Hammerling, Manvitha Ponnapati, Samuel G. Rodriques, and Andrew D. White. Language agents achieve superhuman synthesis of scientific knowledge. *arXiv preprint arXiv:2409.13740*, 2024.

[8] Yijia Shao, Yucheng Jiang, Theodore A. Kanell, Peter Xu, Omar Khattab, and Monica S. Lam. Assisting in writing Wikipedia-like articles from scratch with large language models. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

[9] Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[10] Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. Enabling large language models to generate text with citations. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[11] Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[12] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2023.

[13] Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research (TMLR)*, 2024.

[14] Mo Li, Zixin Yin, Ting Cao, and Yunxin Liu. Training skills like parameters via self-supervised semantic diffusion. *arXiv preprint arXiv:2607.27557*, 2026.

[15] Jingguang Li, Yebo Wu, Zuyi Guo, Kailang Ma, Xianjie Dai, Han Zheng, Benwang Chen, Li Li, Can Rong, and Heye Huang. CoEM: Empowering long-context reasoning with commit-on-evidence memory. *arXiv preprint arXiv:2609.36935*, 2026.

[16] Sean Hardesty Lewis, Zuyi Guo, Benwang Chen, Zirui Liu, Hongyi Lin, and Heye Huang. SafeVantage: Vantage-aware memory for reliable embodied decisions. *arXiv preprint arXiv:2609.36906*, 2026.

[17] Jie Ma, Zhipeng Qian, Yufei Ma, Zihan Liang, Jiayi Ji, Qingpeng Cai, Ben Chen, Peng Jiang, and Xiaoshuai Sun. Skill-V: Verifiable self-evolving skill library for interactive agents. *arXiv preprint arXiv:2610.11781*, 2026.

[18] Bo Tang, Yang Zhang, Guomian Zhuang, Wenqiang Wei, Gaoyang Zheng, Lindong Xie, Yanchao Tan, Feiyu Xiong, Qingyu Yang, Edward Chung, and Zhiyu Li. From memory to skills: Evidence-grounded co-evolution governance for long-horizon LLM agents. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, 2026.

[19] Haozhen Zhang, Quanyu Long, Jianzhu Bao, Tao Feng, Weizhi Zhang, Haodong Yue, and Wenya Wang. MemSkill: Learning and evolving memory skills for self-evolving agents. *arXiv preprint arXiv:2602.02474*, 2026.

[20] Haiming Tang, Xianjie Dai, Gujie Shao, Zuyi Guo, Jingguang Li, Kailang Ma, Yihong Tang, and Heye Huang. MemTransfer: Benchmarking memory beyond matched experience in embodied decision-making. *arXiv preprint arXiv:2609.32313*, 2026.